THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING

DEPARTMENT OF CIVIL ENGINEERING

CIVL1113: ENGINEERING MECHANICS AND MATERIALS

December 12th, 2023 (Tuesday) 2:30 PM – 5:30 PM (3 hours)

Answer any **5 (FIVE)** questions from Section A and B.
All questions carry equal marks.
Marks for sub-questions (if any) are given.

Use a separate answer book for each section.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

SECTION A

1. (a) A simply supported beam ABCD is subject to a vertical upward of 4.0kN at point B and a uniformly distributed load of 2.0kN/m from point C to D as shown in Figure 1(a). The dimension of beam and the load distribution are shown in Figure 1(a).

   (i) Draw the shear force diagram (SFD) with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. **(5 marks)**

   (ii) Draw the bending moment diagram (BDM) with pertinent values. Noe that bending moment should be drawn on the **tensile** side. **(5 marks)**

   (b) A plane frame as shown in Figure 1(b) consists of column AC and beam CD. The column is subject to a horizontal force of 2.0 kN applied at point B and the beam CD is subject to a vertical uniformly distributed load of 1.0 kN/m. Column ABC has a moment release just below joint C.

   (i) Draw the shear force diagram (SFD) with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. **(5 marks)**

   (ii) Draw the bending moment diagram (BDM) with pertinent values. Noe that bending moment should be drawn on the **tensile** side. **(5 marks)**

   Hint: Because of the moment release (also called hinge connection), the moment at joint C must be 0.

2. (a) A cantilever beam ABC is loaded with a uniformly distributed load of 4kN/m at region BC and an external horizontal load at point C of 100kN. Total length of beam is 2 m. The boundary at point A is a fixed boundary. The dimension of beam and the load distribution are shown in Figure 2(a). The cross-section of the beam is shown in Figure 2(b). Neglect self-weight of the beam. Neglect buckling in this problem. The allowable yield stress is 250 MPa after considering safety factor. Assume von-Mises criterion is used. The heigh of beam is h=500 mm. Width of beam is b=100 mm. Top flange and bottom flange are 10 mm. Web plate thickness is 5 mm.

   (i) Find the maximum tensile normal stress at section A. Find the maximum shear stress at section A. **(5 marks)**

   (ii) Check the safety of section A at the connection between top flange plat and web using von-Mises criterion. Check safety of material at section A. **(5 marks)**

   Hint: It is acceptable to assume the shear stress are uniformly distributed in the web plate. During calculation of the moment of inertia of this section as shown in Figure 2(b), it is possible to use the following formula: $I=\frac{1}{12}bh^3-\frac{1}{12}(b-t_w)(h-2t_f)^3$. Where b is the total width, h is the total height, t_w is the web thickness, t_f is the flange plate thickness. When the point is only subject to normal stress σ and shear stress τ, von-Mises criterion is formulated as: $\sqrt{\sigma^2+3\tau^2}\le[\sigma]$

(b) An element shown in Figure 2(c) is in plane stress state. The detailed value and direction are shown in Figure 2(c). The shear stress is 60 MPa. In horizontal direction, the **compressive** normal stress has magnitude of 80 MPa. In vertical direction, the **tensile** normal stress has magnitude of 100 MPa. Assume Tresca criterion is used. The allowable yield stress from uniaxial tensile test is 120 MPa.

(i) Obtain the value of **maximum principal stress** and **minimum principal stress** at that location. **(5 marks)**

(ii) Check the strength condition according to **Tresca criterion**. **(5 marks)**

Hint: Tresca criterion is also called maximum shear stress criterion.

3. (a) A cantilever shaft is fixed at point A and subject to external torsional load of T=10 kN×m at point B as shown in Figure 3(a). The shaft has solid round section with diameter of 100 mm as shown in Figure 3(b). The shear modulus is 80GPa. The total length is 2000 mm.

(i) Calculate the maximum shear stress of the shaft at point A. **(5 marks)**

(ii) Calculate the angle of twist at point C in radians. **(5 marks)**

(b) A long, slender column ABC is pin supported at the ends and compressed by an axial load P as shown in Figure 3(c). Lateral supports are provided at midpoint B. Only consider buckling in the plane of the figure. The column is constructed by a round solid section with elastic modulus of 200 GPa. The total height of column ABC is 10 m. Both AB and BC have length of 5m. The diameter is 100 mm for the column round circular section is as shown in Figure 3(d).

(i) Determine the allowable load P_{allow} using a factor of safety n=2.0 with respect to Euler buckling of the column. **(5 marks)**

(ii) Determine the axial shortening of column at point C under allowable load P_{allow} obtained from the question above. **(5 marks)**

Hint for 3(a) and 3(b): The polar moment of inertia of solid round section is $\frac{\pi d^4}{32}$, where d is the diameter of cross-section. The moment of inertia of solid round section with respect to the neutral axis is $\frac{\pi d^4}{64}$, where d is the diameter of the round section. According to Euler's formula, the critical load for a slender column with span L is:

$$P_{cr} = \frac{\pi^2 EI}{(\mu l)^2} \quad \text{(Euler Formula)}$$

The coefficient μ is 1.0 for both pinned ends, μ is 2.0 for one fixed end and one free end, μ is 0.7 for one fixed end and one pinned end, and μ is 0.5 for both fixed ends.

FIGURES

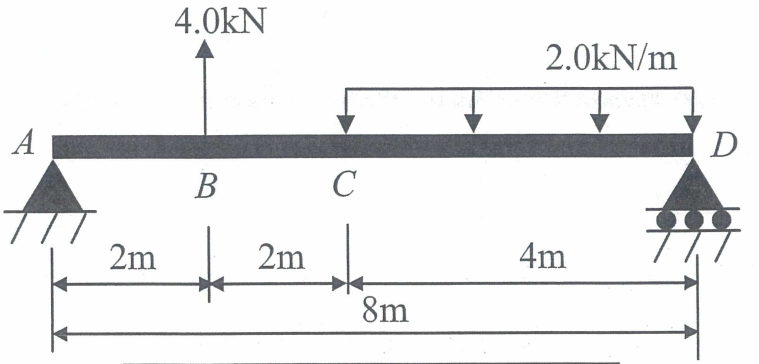


Figure 1(a). Beam

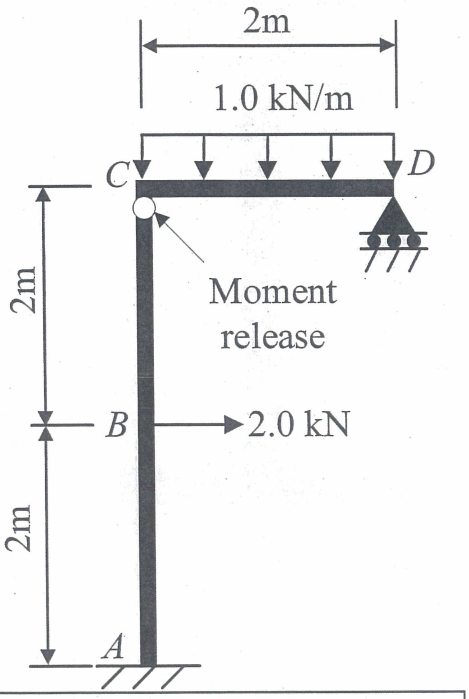


Figure 1(b). Frame

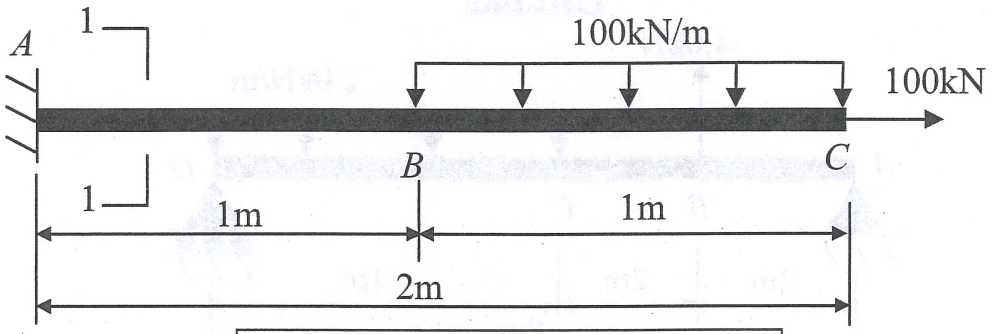


Figure 2(a). Cantilever beam

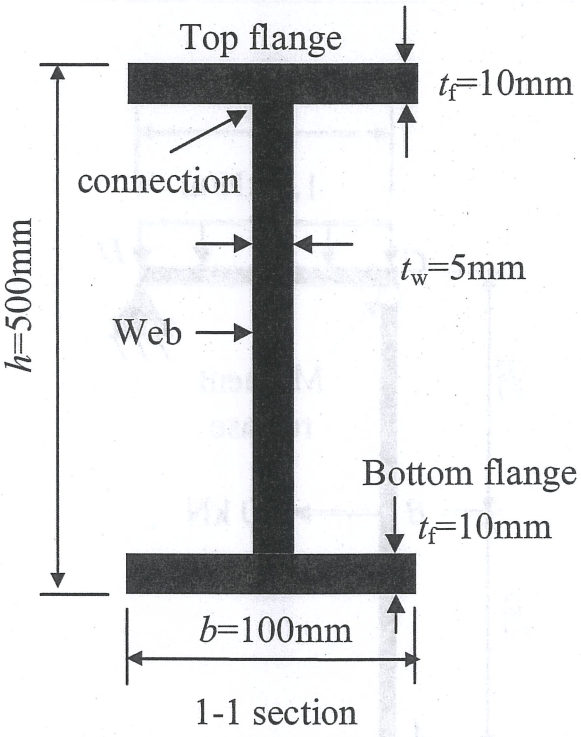


Figure 2(b). Cross section of beam

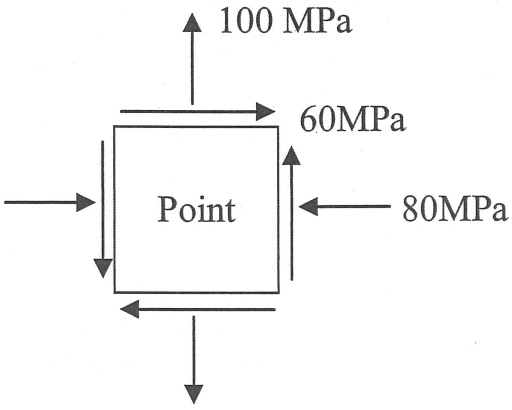


Figure 2(c). Stress state at a point

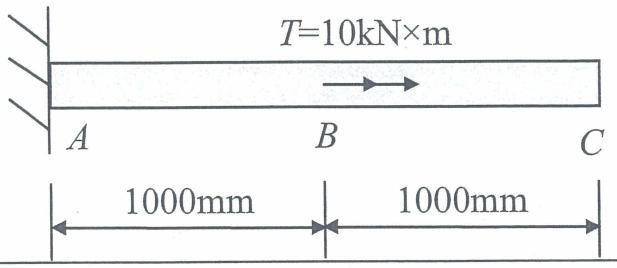


Figure 3(a). Shaft under partial torsion loading

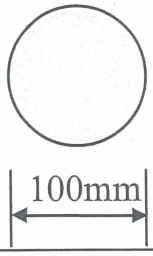


Figure 3(b). Cross section of shaft

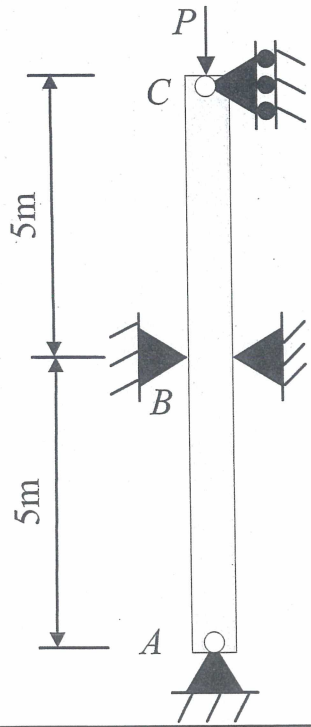


Figure 3(c). Truss

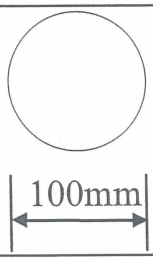


Figure 3(d). Cross section of truss

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